ModelIt!

Teacher's Guide

Can a River Change Direction?

How Flowing Water Reshapes the Land Over Time

Quick Reference

Grade Level	Grade 05L2
NGSS Standard	5-ESS2-1, 5-ESS2-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 5-ESS2-1, 5-ESS2-2

Develop a model using an example to describe ways the geosphere, hydrosphere, atmosphere, and biosphere interact. Describe and graph the amounts and percentages of water and fresh water in various reservoirs to provide evidence about the distribution of water on Earth.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model showing how water flow and soil properties control erosion, sediment transport, and channel change.
Disciplinary Core Idea	**ESS2.A Earth Materials and Systems** — Students model how water interacts with earth materials to reshape landforms through erosion and deposition.
Crosscutting Concept	**Stability and Change** — Students explore how river systems appear stable but are constantly changing through slow erosion processes.

CAST Testing Alignment

MULTIPLE CHOICE:

A river flows through an area where one bank is soft clay and the other is solid rock. Over time, which direction will the river shift and why?

CONSTRUCTED RESPONSE:

Explain how a river can change direction over time using the concepts of erosion, sediment transport, and deposition from your model.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Rivers are powerful agents of change on Earth's surface. Over time, flowing water reshapes the land by eroding material from some areas and depositing it in others. This constant process of erosion and deposition is how rivers carve valleys, create floodplains, and even change their own course.

Erosion: Water's Sculpting Tool

Erosion occurs when flowing water has enough energy to dislodge and carry away soil and rock particles. Faster water has more erosive power. On the outside of a river bend, water flows fastest and cuts into the bank.

Sediment Transport

Once material is eroded, the river carries it as sediment. Faster water can carry larger and heavier particles. When water slows down — on the inside of a bend or where the river widens — it drops its sediment load in a process called deposition.

How Rivers Change Course

Rivers develop curves called meanders because erosion on the outside of bends and deposition on the inside cause bends to grow larger over time. Eventually, a meander can loop so far that the river cuts a new, shorter path, leaving an oxbow lake behind.

Factors That Control Erosion

The rate of erosion depends on water speed, soil or rock type, vegetation cover, and slope. Soft soils like clay and sand erode quickly, while hard rock like granite resists erosion for thousands of years.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Water Flow Speed, Soil Softness, Erosion Rate, Sediment Transport, and Channel Shape Change as the five system components.

E **Establish Relationships**

Students determine that Water Flow Speed increases Erosion Rate (+), Soil Softness increases Erosion Rate (+), Erosion Rate increases Sediment Transport (+), Sediment Transport increases Channel Shape Change (+), and Water Flow Speed increases Sediment Transport (+).

V **Visualize & Model**

Students build the river erosion model showing how external conditions (water speed and soil type) drive internal responses (erosion, transport, channel change).

E **Evaluate Outcomes**

Students run the Fast River, Hard Rock scenario and discover that even fast water erodes slowly through hard rock, then compare with soft soil scenarios to see the dramatic difference.

R **Revise & Extend**

Students add Vegetation Cover or Rainfall Intensity to explore how plants and weather affect river erosion.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Can a river actually change direction? The answer might surprise you."*

Watch For

Show satellite images of meandering rivers and ask: Has this river always looked like this?

Slide 2: Learning Objectives

2 min

Say: *"Today we model how water reshapes the very ground it flows over."*

Watch For

Students read objectives and share what they think 'erosion' means.

Slide 3: Big Question

2 min

Say: *"Rivers seem permanent, but they are constantly moving. How?"*

Watch For

Partner discussion: Have you ever seen evidence of erosion?

Slide 4: LEVER Framework

2 min

Say: *"Our model shows how TWO external factors combine to reshape a river over time."*

Watch For

Preview: water speed and soil type work together.

Slide 5: Activity 1: Components

8 min

Say: *"Water Flow Speed and Soil Softness are external — we control them. Erosion, sediment, and channel change respond."*

Watch For

Students sort component cards into external and internal categories.

Slide 6: Activity 2: Connections

8 min

Say: *"Notice that ALL relationships here are positive. More speed means more erosion, more erosion means more transport, and more transport means more channel change."*

Watch For

Guide students to draw all five relationship arrows in ModelIt.

Slide 7: Activity 3: Simulation

10 min

Say: *"Fast water in soft soil vs fast water in hard rock — what is the difference? Let's test it."*

Watch For

Students run all three scenarios and record observations.

Slide 8: Discoveries

5 min

Say: *"Rivers are not permanent. They are constantly reshaping themselves and the land around them."*

Watch For

Class discussion: What does this mean for communities near rivers?

Slide 9: STEM Challenge

10 min

Say: *"A community needs your help to save their riverbank. Design a protection plan."*

Watch For

Groups brainstorm, design, and present their protection strategies.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Water Flow Speed, Soil Softness
- **INTERNAL:** Erosion Rate, Sediment Transport, Channel Shape Change

Relationships

- **Water Flow Speed to Erosion Rate:** POSITIVE (+) — Faster water has more energy to dislodge and carry away soil and rock particles.
- **Soil Softness to Erosion Rate:** POSITIVE (+) — Softer soil is easier for water to break apart and carry away.
- **Erosion Rate to Sediment Transport:** POSITIVE (+) — More erosion produces more loose material for the river to carry.
- **Water Flow Speed to Sediment Transport:** POSITIVE (+) — Faster water can carry more and larger sediment particles.
- **Sediment Transport to Channel Shape Change:** POSITIVE (+) — More sediment being moved and deposited causes greater changes in the river's path and shape.

Key Discoveries

- Rivers CAN change direction over time through erosion on the outside of bends and deposition on the inside
- Faster water erodes more material and carries more sediment, reshaping the channel quickly
- Soft soil erodes much faster than hard rock, so rivers change course more easily in soft ground
- Channel shape change is a SLOW process — it happens over years, decades, and centuries, not overnight

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Rivers have always been where they are now"	Rivers constantly change course through erosion and deposition. The Mississippi River has shifted hundreds of miles over thousands of years. Even small streams shift noticeably over decades.
"Erosion only happens during floods"	Erosion happens every day as water flows past riverbanks. Floods accelerate erosion dramatically, but normal flow constantly wears away soft material. It is a slow, continuous process.
"Rocks and soil are permanent"	Rocks and soil seem permanent because erosion is usually too slow to watch. But given enough time, water can carve through solid rock — that is how the Grand Canyon formed over millions of years.